E-COMMERCE SALES DATA ANALYSIS AND DASHBOARD

1. Introduction

This project focuses on analysing sales data using multiple datasets, including Customer, Product, Store, and Sales Fact. The purpose of this analysis is to understand sales performance, identify trends, and generate meaningful insights that can support business decision-making.

2. Objectives

- To analyse sales data using multiple datasets such as Customer, Product, Store, and Sales Fact
- To clean and pre-process the data to ensure accuracy and consistency
- To integrate different datasets using common keys (Customer ID, Product ID, Store)
- To extract meaningful information using formulas like XLOOKUP and IF
- To calculate key metrics such as Total Sales
- To summarize the data using Pivot Tables
- To visualize sales performance using charts and graphs
- To identify top-performing products, cities, and customers
- To generate insights that support better business decision-making

3. Dataset Description

The project uses four datasets:

- Customer Dataset: Contains customer-related information such as Customer ID and Customer Name
- Product Dataset: Contains product details such as product ID and Product Name
- Store Dataset: Contains store-related information such as Store ID and City
- Sales Fact Dataset: Contains transactional data including Customer ID, product ID, Store ID, Quantity, and Price

4. Data Cleaning & Preparation

Data cleaning was performed to ensure the quality and accuracy of the datasets. The following steps were carried out:

- Removed duplicate records
- Handled missing values
- Corrected data formats (date, number, text)
- Ensured consistency across all datasets
- Additionally, Power Query was used (if applicable) to efficiently clean and transform the data.

5. Data Integration

The datasets were integrated using common keys:

- Customer ID was used to connect with the Customer dataset
- product ID was used to connect with the Product dataset
- Store ID was used to connect with the Store dataset
- XLOOKUP function was used to retrieve relevant fields such as Customer Name, Product Name, and Store City into the Sales Fact table.

6. Data Processing

After integration, additional calculations were performed:

- Total Sales was calculated using the formula: Quantity × Price
- An IF condition was used to categorize sales as High or Low

These transformations helped prepare the data for further analysis.

7. Analysis (Pivot Tables)

Pivot tables were created to summarize the data:

- Product-wise sales analysis to identify top-selling products`
- Customer-wise sales analysis to identify key customers

8. Data Visualization

Charts were created to represent the data visually:

- Bar charts for comparing product sales
- Pie charts for showing distribution
- Column charts for overall performance

A dashboard was created to present all visualizations in a single view.

Dashboard Creation

9. Key Insights

- Sales are not evenly distributed and are concentrated among specific customers, brands, and regions.
- The analysis helps in identifying high-performing segments and areas with growth potential.

10. Conclusion

This project provided a clear understanding of sales performance through data analysis. By integrating and analysing multiple datasets, valuable insights were generated. These insights can help businesses make informed decisions and improve overall performance.